CoffeeBot Platform: Green Computing Justification

Prepared 10 November 2026 | Measurements: Lighthouse (local production build) and Website Carbon Calculator (coffeebot-platform.vercel.app)

Summary

The CoffeeBot platform is green in two separate ways. It is **green by purpose**: it exists to keep used coffee grounds out of landfill and into reuse. It is also **green by design**: it is built as a lightweight static web app that produces very little carbon per visit.

> *"Our web app keeps coffee waste out of landfill, and it is built so lightly that a visit produces only 0.07 g of CO2, which is cleaner than 88% of web pages globally."*

1. Green by purpose (what the app does)

Used coffee grounds sent to landfill rot and release methane, a strong greenhouse gas. The platform is a digital Circular Sustainability Hub that works against this.

- **Measures the loop:** staff dashboards log kilograms of grounds collected, reused and disposed, by partner and by use (compost, mushroom growing, activity kits, landfill).
- **Connects the reuse chain:** schools, workplaces and reuse partners (cafes, farms, makers) book activities and send requests to take grounds.
- **Changes behaviour:** the Education Hub, quizzes and home missions, shared by link, teach students and families about reuse.
- **Makes impact reportable:** downloadable reports support sustainability and ESG accountability.
- **Aligns with global goals:** circular economy principles, UN SDG 12 (responsible consumption and production) and SDG 13 (climate action).

2. Green by design (how the app is built)

- **Static site, no dedicated application server:** hosted on Vercel with state held in the browser, so there is no always-on backend consuming energy.
- **Route-level code splitting:** most pages are under 10 kB gzipped, so visitors download only what they use.
- **Lazy loading:** the GSAP scroll library loads on demand and on desktop only; Supabase loads after first paint.
- **Respects reduced motion and mobile:** the animated landing page falls back to simple stacked sections, saving device CPU.
- **Efficient assets:** WebP imagery, self-hosted subset woff2 fonts, SVG favicon.
- **Light data sync:** shared-state writes to Supabase are debounced rather than constantly polled.

3. Evidence (measured)

Website Carbon Calculator (tested 10 Nov 2026)

Metric	Result
Carbon rating	**A**

Metric	Result
CO2 per page view	**0.07 g**
Compared with all web pages	Cleaner than **88%** of pages globally
At 10,000 page views per month	8.11 kg CO2e per year (about what 1 tree absorbs in a year), 16 kWh
Hosting	Not on the green host list; green hosting would cut a further 9%

Lighthouse audit (landing page, local production build)

Metric	Result
Performance score	**87 / 100**
Best practices score	**100 / 100**
First Contentful Paint	1.6 s
Largest Contentful Paint	4.0 s
Total Blocking Time	90 ms
Cumulative Layout Shift	0.001
Landing page first-load weight	575 KiB across 16 requests
All JavaScript (all routes)	851 kB raw, 282 kB gzipped
All CSS	71 kB raw, 21 kB gzipped

4. What the tools are

Lighthouse is a free audit tool from Google, built into Chrome. It loads a page in a controlled test and scores speed and build quality out of 100 (90 and above is good). It is not a carbon tool; it shows the site is efficiently built, and an efficient page does less work on servers, networks and devices.

Website Carbon Calculator (websitecarbon.com, by Wholegrain Digital) estimates the CO2 produced per visit. It measures the data a page transfers, converts it to energy using the Sustainable Web Design model (data centre, network and device), converts energy to grams of CO2 using average grid intensity, and checks whether the host is a verified green host. Ratings run from A+ (best) to F.

5. Honest limitations

- The hosting is not certified green (about 9% further saving possible).
- Website Carbon tests one page (the landing page), not the logged-in dashboards, and gives an approximation based on averages.
- Three PNG icons (scrub, grounds, cup) total about 317 KB, roughly 54% of the landing page weight. Converting them to WebP or SVG is the clearest improvement and would also help the 4.0 s Largest Contentful Paint.
- The kilogram figures shown in the app are currently demo (seed) data, so real-world tonnage should not be claimed yet.
- The app's footprint and its sustainability benefit are separate arguments: the benefit comes from the reuse the app enables, not from the code itself.

6. Possible next steps

- Convert the three PNG icons to WebP/SVG and re-test.
- Add long-lived cache headers for /assets and /fonts in vercel.json.
- Lazy-load below-the-fold images.
- Move to, or confirm, a Green Web Foundation verified host.
- Add a CO2e estimate for grounds diverted from landfill to the staff dashboard.

7. Answers to likely questions

Isn't every website bad for the planet? Every site uses some energy, which is why this one was measured. Its footprint is very low, and its purpose (less landfill methane) outweighs it.

Is the sustainability real or marketing? The platform tracks grounds per batch and partner. Current figures are demo data, which should be stated plainly.